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Prediction of Air Pollution Index in Kuala Lumpur using fuzzy time series and statistical models

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Abstract
Air pollutants can cause multifaceted harm to the human body. Respiratory diseases and immunology dysfunction are some of its main manifestations. Forecasting the air quality of a country is important to allow the government to take preventive measure. In this research, the artificial neural network (ANN), autoregressive integrated moving average (ARIMA), trigonometric regressors, Box-Cox transformation, ARMA errors, trend and seasonality (TBATS) and several fuzzy time series (FTS) models are utilized in the forecasting of air pollution index (API) of Kuala Lumpur, Malaysia, for the year 2017. Six years of daily API data for Kuala Lumpur from the year 2012 to the year 2017 for the Cheras observation station in Kuala Lumpur has been selected as the dataset of this research. The mean absolute percentage error (MAPE), root mean square error (RMSE) and computational time have been used as the performance evaluation metrics for the models and these values were calculated for each of the chosen forecasting models. A brief but comprehensive comparative study of the results obtained from each of the chosen model is presented in order to identify the most effective model to forecast API values. It was found that the fuzzy time series models outperformed the other models in terms of accuracy of forecasted values and computation time. Specifically, the Singh fuzzy time series model was found to be the most accurate and efficient forecasting model with RMSE of 1.4704 and MAPE of 4.364%.

Keywords Air Pollution Index (API) · Artificial neural network (ANN) · Autoregressive integrated moving average (ARIMA) · Trigonometric regressors · Box-Cox transformation · ARMA errors · Trend and seasonality (TBATS) model · Fuzzy logic · Fuzzy time series

Introduction

Air pollution is a type of pollution caused by the substances that endanger the human body and the environment. The substances that caused air pollution can be gases, liquid suspensions or solid form. The main substances that caused air pollution are smoke particles, dust, domestic garbage and automobile exhaust (Choudhary and Garg 2013). Sources of air pollution can be classified into sources of natural pollution and human-made pollution. Natural pollution sources are formed by natural causes such as eruptions of volcanos, sea salt particles, or forest fire. On the other hand, sources of anthropogenic pollution are formed by the activities of people. Among the sources of human-made pollution, it can be further classified into fixed, such as industrial smokestacks, and mobile which are pollutants from cars, trains, aeroplanes and ships. Man-made pollution sources are ubiquitous, and

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therefore, they should be more closely watched than the natural sources of pollution.

Air Pollution Index is formerly used in Hong Kong; however, it is then replaced with Health-Based Index (HBI) (Wong et al. 2012). API is developed after the Pollutant Standard Index (PSI) which was presented by the United States Environmental Protection Agency (US-EPA) and it is a simple method that is generalized in predicting the air quality in Malaysia (Malaysia Department of Environment 2019). It allows the public to easily understand the status of current air quality. The calculation of API is based on the density of five main pollutants, which are the sulphur dioxide (SO_2), particulate matter with a diameter less than 10 μm (10), ozone (^{3}), carbon monoxide (CO_2) and nitrogen dioxide (NO_2) (Marković et al. 2007). The pollutant was calculated based on the standard listed by the WHO (World Health Organization 2005). Different sub-index values of API indicate different health impacts. The highest sub-index among the five main pollutants is chosen as the API for a specific period.

The air is made up of various chemical compositions, which the most prevalent element is nitrogen, followed by oxygen and other gases. The composition of each gas is different every day. The small number of pollutants in the air only gives minor effect to the human body and the surrounding environment; however, when the concentration of pollutants in the air reaches the harmful levels, action should be taken to reduce the detrimental pollutants. Therefore, it is important to constantly evaluate and forecast air quality in a country. The methods and regulations to determine the air quality vary from country to country and the terms used in each country are slightly different. The Air Pollution Index (API) is used in Malaysia to determine the current condition of the air quality and therefore proper actions can be taken to minimize the impact of air pollution to the people and environment.

There are a wide variety of techniques to forecast air quality and there are many existing works related to the study of air quality in literature. Pereira et al. (2019) presented a comparative study of PM10 in the three Latin American cities of Lima, Medellin and Sao Paulo. Sharma et al. (2019) on the other hand presented a study on inferring the level of air pollution in different regions in India using data collected from the major cities of Bengaluru, Delhi and Chennai in India. Elsharkawy and Ibrahim (2019) studied the sources and concentrations of acidic constituents in the air using data from Dammam City in Saudi Arabia, whereas Mirzaei et al. (2019) investigated the spatial and temporal variability relationship between PM2.5 concentrations using data collected from ground-monitoring stations and satellite data in Tehran. Lei et al. (2019) used several statistical models to forecast air quality in Macao for the year 2017, whereas Wang et al. (2019) investigated the use of ANN in the online monitoring of air pollutants using data obtained from eight air quality–monitoring stations in Shanghai, China. Agudelo-Castaneda et al. (2014) studied the concentration of surface ozone and nitrogen oxides in an urban area in Brazil over a 4-year period using the Kolmogorov-Zurbenko filter. Cujia et al. (2019) forecasted the concentration of PM10 using 16 years of data collected from air quality–monitoring stations from several Caribbean cities along the northern coast of Colombia, while Wang et al. (2019) used a fuzzy time series early warning system for urban air quality forecasting in China. Freeman et al. (2018) used a deep learning model that consists of a recurrent neural network and (RNN) and long short-term memory (LSTM) model to forecast air quality. Zhu et al. (2018) on the other hand introduced a new optimal forecasting model which was made up of 3 forecasting models and used this to forecast air quality index in 5 cities in North China. In this research, ANN, ARIMA, TBATS and FTS are used to forecast the API values in Malaysia.

Artificial Neural Network (ANN) is a complex network with many simple mechanisms linked together, with strong nonlinearity, a system capable of complex logical operations and nonlinear relationship. It serves as a nonlinear time series–forecasting method with an unspecified model term.

Autoregressive integrated moving average (ARIMA) model is commonly used in the modelling of the time series as it can present both the nonstationary and stationary time series. Besides, fitting ARIMA into time series can make the data easy to be comprehended or make use of it for the prediction of future values. De Livera et al. (2010) have introduced the TBATS model which incorporates the Box-Cox transformations, ARMA error correction and Fourier series with time-dependent coefficients to model and forecast complex seasonal time series. This type of complex time series usually includes multiple seasonal periods, noninteger seasonality, high-frequency seasonality, and dual-calendar effects that cannot be modelled effectively by conventional time series forecasting models. TBATS model includes likelihood evaluation and trigonometric formulation, which provides a simple but comprehensible method to predict complex seasonal time series. Also, the trigonometric decomposition in the model can examine and extract the seasonal elements that are less obvious in the time series.

The theory of fuzzy sets was introduced by Zadeh in 1965. One of the concepts of fuzzy logic (FL) that plays a significant role in applications is the IF-THEN rules or simply known as fuzzy rules. FL has been heavily used in areas such as automation control, diagnosis and forecasting. Besides, FL can greatly facilitate the process of solving various problems, building expert systems and neural networks.

Preliminaries

In this section, we present a brief study of the concepts that are pertinent to the development of this paper.

Artificial neural network

McCulloch and Pitts (1990) introduced the method of ANN, a computational model that imitates the working of a biological brain network. It is a nonlinear time series–forecasting method, with an unspecified model form. This means that the ANN does not have a prior assumption on the model structure like the other modelling techniques; yet, it obtains the structure by generalizing the training data. This characteristic of ANN offers itself a more flexible framework than other statistical forecasting techniques to capture nonlinear time series data (Zhang 2003). The ANN can be trained using training data to recognize nonlinear patterns between input and output values (Rahman et al. 2013). It can be trained more precisely when added with new data, and due to its generalization properties, the weights of the links can be formed by training. This can then be inserted into existing technology, helping in solving difficult problems a lot quicker than the computers (García et al. 2011). Based on the characteristics of the ANN, a lot of researches have applied it, particularly the multilayer perceptron (MLP) models, in predicting the environmental situation (Roadknight et al. 1997; Bai et al. 2018). The ANN is an effective tool for modelling environmental systems, especially being used to forecast air quality (Maier and Dandy 2001). However, according to Heo and Kim (2004), the forecast accuracy of ANN might be affected if previously unknown data patterns are included in the training data. Also, according to Morabito and Versaci (2003), the downside of ANN is that the knowledge gained about an issue is encoded in the connections' weights which require users to further fix the strengths of the weights after the training of the data.

ARIMA

The ARIMA model is also known as the Box-Jenkins model. This model uses a few parameters to explain the events or the process which are generated by the random walk. ARIMA models can present both the nonstationary and stationary time series. There are three main steps that require to be taken into considerations. These steps are parameter estimation, tentative identification and diagnostic checking when the model is built for as a predicting model (Hanke and Wichern 2005). The ARIMA model is typically expressed as $ARIMA(p, d, q)$ and the parameter indicate the specific model that is used in the research. The parameter of the ARIMA models is defined as follows:

- 2022p represents the count of lag observations used in the ARIMA model. It captures on the autoregressive of the ARIMA models.
- d represents the count of times that the raw data is differentiated in this model to produce a stationary time series.
- q represents the size or the order of moving average of the model.

TBATS

The TBATS model utilizes the roots in the exponential smoothing methods and incorporates likelihood evaluation and trigonometric formulation to identify the complex seasonality in the time series. The TBATS model uses a state space model, which is a generalization of those underpinning exponential smoothing. In TBATS, the seasonality is modelled using the trigonometric representation based on the Fourier series. The two major advantages of the TBATS model is that it only needs two seed states regardless of the length of period, as well as its ability to model the seasonality of noninteger period.

Fuzzy time series

Fuzzy time series was first introduced by Song and Chissom (1993) who had initiated the use of fuzzy time series (FTS) to forecast scenarios with these characteristics: (i) dynamic in process; (ii) observations fulfil the characteristics of fuzzy sets; (iii) the fuzzy set's domain of discourse is a subclass of R as defined by Zadeh (1973); (iv) the procedure is unable to be described by the normal time series models. Hence, the FTS is suitable to be applied for the decision-making process in a complex system where the circumstance of the issue is often not clear. The characteristics of membership function in the fuzzy sets have made it possible for past researchers such as Fisher (2003) to apply a fuzzy set in describing air pollution levels as air quality can be categorized into good, medium or poor quality.

Several important definitions of the FTS are outlined below:

- Definition 2.3 (Song and Chissom 1993): Let $F(t)(t = 0, 1, 2, 3, \ldots)$, which is a subset of the domain of discourse, $\mathbb{R}$, denoted by the fuzzy set of $g_i(t)$. If $G(t)$ is the collection of all $g_i(t)$, then $G(t)$ is the FTS on $Z(t)$. According to this case, $G(t)$ is the linguistic variable while $g_i(t)$ is any possible linguistic variable in $Z(t)$.

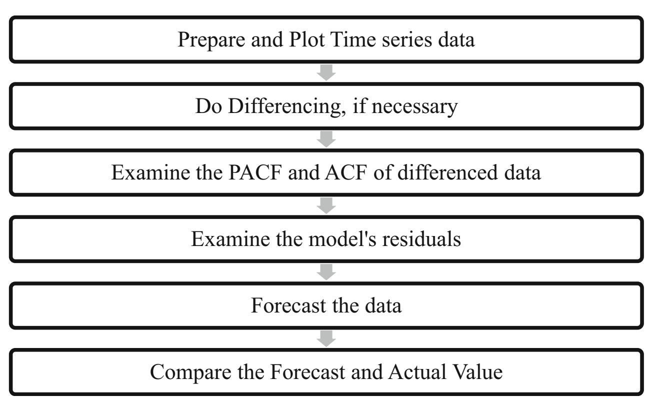

Fig. 1 Procedure of forecasting API using ARIMA

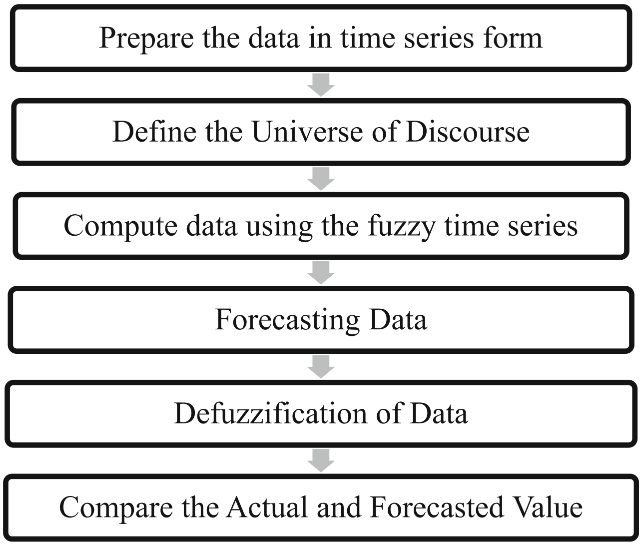

Fig. 2 Forecasting method for the FTS method In this paper, three different FTS models namely Chen (1996) model, the heuristic model by Huarng (2001) and Singh (2008) model are used to forecast the API for the year 2017. The RStudio software is used for the forecasting process using the FTS model

- Definition 2.4 (Song and Chissom 1993): When a fuzzy relation $S(t, t-1)$ exists amid $F(t)$ and $F(t-1)$, where $F(t) = F(t-1) * S(t, t-1)$, while the $*$is an operator of the composition, then $F(t)$ is said to be triggered by $F(t-1)$. Also, $F(t)$ is represented by a fuzzy logical relationship (FLR), written as $F(t-1) \rightarrow F(t)$. Hence, S, the FLR is defined as the first-order model of $F(t)$.

Based on the definition of FTS in Song and Chissom (1993), Cheng et al. (2008), Chen (1996) and Yu (2005) had improved on the fuzzy time series algorithm by using different approaches to establish the fuzzy relationships that are studied.

Rahman et al. (2013) forecasted API using the ANN, FTS and ARIMA models and used 10 years of monthly API readings of the residential and industrial area in Malaysia for this research. The RMSE of the training and testing performance was calculated to make a comparison. The performance of the prediction using ANN methods in the research was more satisfactory compared with the SARIMA model, as well as Chen, Yu and Cheng's fuzzy time series models.

The research done by Bai et al. (2018) remarked that the fuzzy methods are less accurate and unsystematic, though having a great error endurance and not needing precise mathematical models during every computer-human interaction. This is because the membership function of a fuzzy set is difficult to be precisely defined. Hence, expert judgement is needed to assign the range of ambiguity in determining air quality (Fisher 2003). Also, the existence of outliers would affect the forecasting ability of the FTS. The authors therefore remarked that during its application, the fuzzy method could be further optimized using genetic algorithms.

Proposed methodology

Throughout this section, we will discuss the research methods used in this paper. The description of the sampling sites, data source, evaluation metrics and the forecasting procedures will be presented.

Description of the sampling sites

Kuala Lumpur is selected as the sampling site of this study because it is the national capital of Malaysia and the largest city in the country. Kuala Lumpur is located at the west central of Peninsular Malaysia and consists of many residential areas. The population in Kuala Lumpur is increasing rapidly and its total population has increased from 1.74 million in 2014 to 1.8 million in 2018. It is selected as the sampling site due to its large population nature. Tracking the air pollution level in this city is significant as a lot of people's health may get affected if the city's air quality deteriorates. Preventive measures can be done if any forecasting model can obtain an accurate forecast of the API. In this research, the historical API readings in Kuala Lumpur are needed to train and test the forecasting models. The API readings, managed by the Malaysia Department of Environment (DOE), are obtained from the Malaysia Public Sector Open Data Portal. The daily API data from the year 2012 to 2017, collected from the Sekolah Menengah Kebangsaan Seri Permaisuri, Cheras, Kuala Lumpur station, is utilized in this study to reflect the air quality in the city. The dataset from the year 2012 to 2016 are used as

Table 1 The results of the ANN model

Trial	First trial	Second trial	Third trial
Number of hidden layers	1	2	3
Number of neurons for each of the layers	8	8, 3	6, 5, 3
MAPE	78.781%	79.0514%	78.7357%
RMSE	22.21475	22.36465	22.72587
Computational time (min)	2.17	2.32	3.03

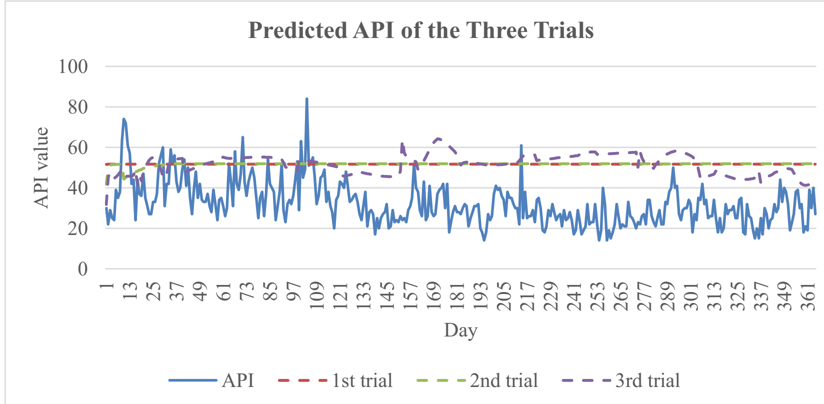

Fig. 3 The trend of the actual API and the forecasted API from ANN model

training set while the 2017 dataset is used to test the accuracy of the forecast results.

Evaluation metrics

The accuracy of the forecasted values is evaluated using the mean absolute percentage error (MAPE) and root mean squared error (RMSE). The formulae to calculate the metrics are shown below:

$$\mathrm{RMSE} = \sqrt{\frac{\left(\mathrm{API}_{\mathrm{obeserved}} - \mathrm{API}_{\mathrm{predicted}}\right)^2}{\text{number of } \mathrm{API}_{\mathrm{obeserved}}}} \quad (1)$$

$$\mathrm{MAPE} = \frac{\left|\mathrm{API}_{\mathrm{obeserved}} - \mathrm{API}_{\mathrm{predicted}}\right| / \mathrm{API}_{\mathrm{observed}}}{\text{number of } \mathrm{API}_{\mathrm{obeserved}}} \times 100\% \quad (2)$$

The computational time taken by each model during the forecasting stage was recorded as well.

Forecasting procedures

Ann

The package ‘neuralnet’ in RStudio software is utilized for artificial neural network modelling. The neuralnet package provides a focused function on multi-layer perceptron (MLP) and is useful to model the functional relationship of ANN. It is used to train the data and recognize the nonlinear pattern between the input and output layer. The API values obtained from the DOE is normalized and set as the input layer of the neuralnet.

Step 1: Set train and test data There is a total of 6 years of daily API readings collected in this research. The dataset is divided into two parts. The first 5 years which is from the year

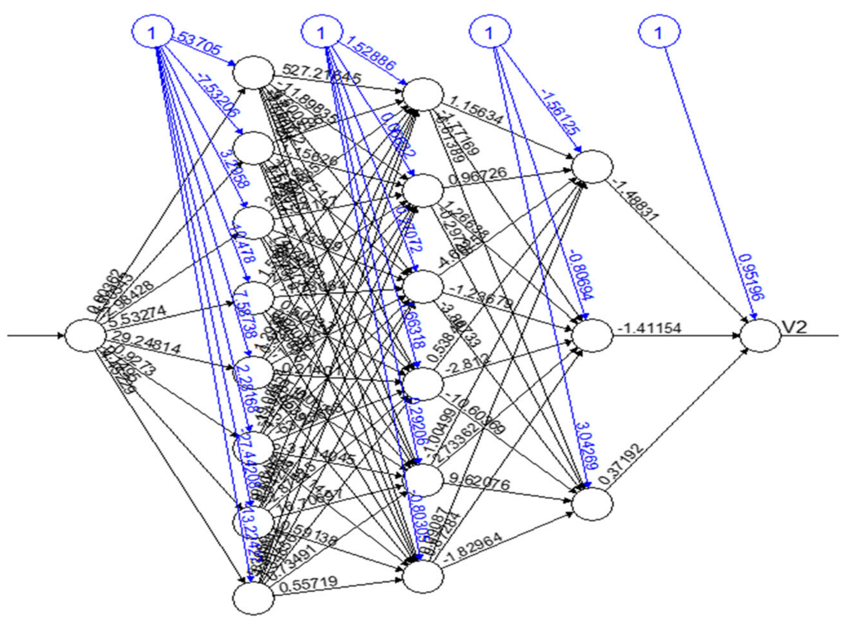

Fig. 4 The neural network of the problem

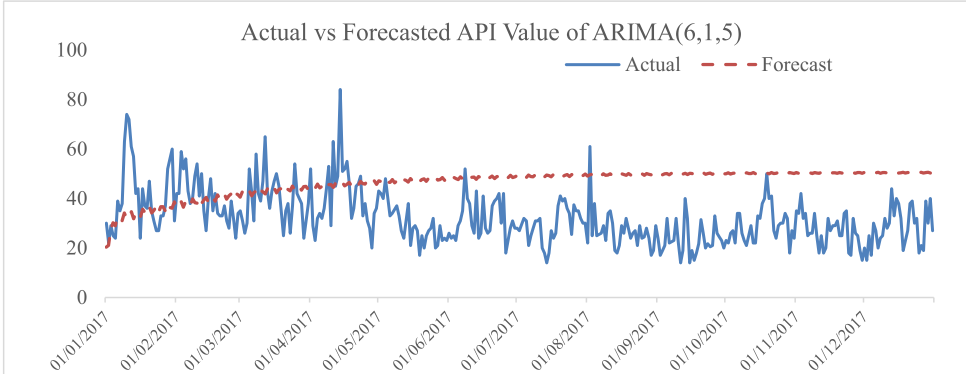

Fig. 5 Actual API vs forecasted API from ARIMA model

2012 until 2016 of API readings in Kuala Lumpur are set as the training data of ANN. The final year of the API readings which is the year 2017 is set as the test data for ANN.

Step 2: Normalize the API readings Normalizing of data is needed before fitting the data into the neuralnet package. Normalizing of data is needed to reduce the difference in the training results. The API readings collected in this research are normalized by subtracting the minimum API readings of the data and dividing it by the difference of the minimum and maximum API readings. The formula to normalize the data is given as follow:

$$\text{Normalization of data} = \frac{API_{observed} - min_{API}}{max_{API} - min_{API.}}$$

Step 3: Set the normalized data The normalized data is set before fitting into the neuralnet same as in step 1. The data of the first 5 years is set as the train data in ANN and the last year of the data is set as the test data in ANN.

Step 4: Fit the data to neuralnet package The date of the data collected is set as the input for the ANN. Certain numbers of layers and neurons are tested to gain higher accuracy in the prediction of API readings.

Step 5: Prediction of API readings The data train in the ANN allowed us to predict the API for the year 2017. The code of 'predict.nn' is used in the RStudio for prediction of the API readings. The predicted API values are then multiplied by the difference of the minimum and maximum API readings of the data, then plus the minimum amount of the API readings.

Step 6: Comparison of the API readings The comparison between the predicted API readings by the ANN models and the test data for the data is done in this step. The graph is plotted to watch the trend of the predicted and the real data readings. The RMSE and MAPE of the predicted values are then calculated using Eqs. (1) and (2), respectively to evaluate the accuracy of the ANN model in predicting the API values. The computational time is also calculated for comparison purpose.

ARIMA model

The following schematic diagram given in Fig. 1 outlines the research procedure for API forecasting using the ARIMA model.

In this paper, we have used the ARIMA(6, 1, 5) model to forecast the API reading for the year 2017. The forecasted result is then plotted against the actual value of the API reading in 2017. The MAPE and RMSE of the predicted data are calculated to measure the accuracy of the ARIMA model in forecasting the API values. The computation was done using RStudio software.

TBATS model

For the TBATS model, the (1, {1,3}, -, {<7,2>, <30,4>, <90,1>, <365,5>}) model is used to forecast the API index for the year 2017. The computation was done using RStudio software.

Fuzzy time series models

The schematic diagram in Fig. 2 outlines the general research procedure for API forecasting using the fuzzy time series methods.

Table 2 MAPE and RMSE for the ARIMA and TBATS model

Forecasting model	ARIMA (6, 1, 5)	TBATS (1, {1,3}, -, {<7,2>, <30,4>, <90,1>, <365,5>})
MAPE	67.595%	43.861%
RMSE	19.89427	15.1563
Computational time (min)	2.02	2.13

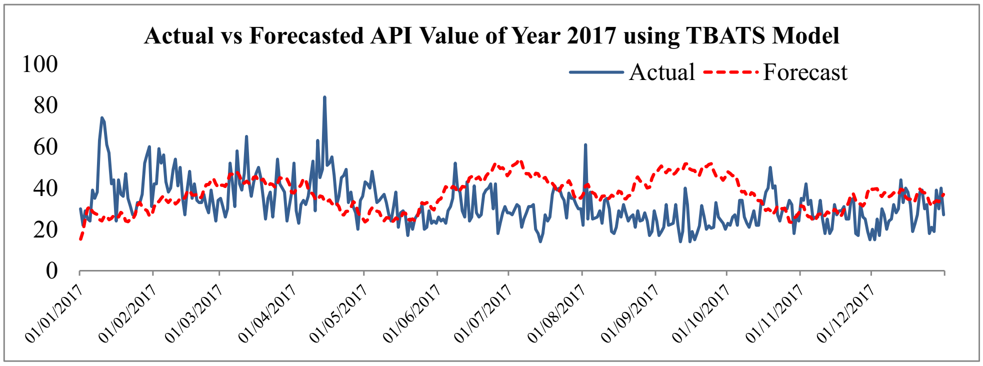

Fig. 6 Actual API vs forecasted API from TBATS model

Performance evaluation

Analysis of results for the ANN model

There is no certain method for selecting the right structure for ANN models. The different number of layers and neurons is determined by trial and error to get the best solution for the prediction of API. The RMSE and MAPE of each of the trial are calculated and the best result for the prediction of API is selected as the result of the ANN and used to compare with the performance of the fuzzy time series models and ARIMA model. In this paper, we have used three trials for the number of layers and neurons for the ANN model. The first trial is one layer of the hidden layer with eight neurons, the second trial is two layers of the hidden layer with six and three neurons respectively and the third trial is three layers of a hidden layer with six neurons in the first hidden layer and five and three neurons respectively in the second and third hidden layers. The RMSE and the MAPE of the prediction of the API of each of the trial are calculated. The computational time of each trial is calculated and showed in Table 1 below.

The values of MAPE showed the mean of the percentage error of each of the predicted API. The lower the MAPE, the better the performance of prediction. Besides, the RMSE is used to measure the difference between the predicted API readings and the actual API readings. A lower value of RMSE indicates better predicted results for the API. From Table 1, it can be seen that the first trial with one hidden layer provides the lowest value of MAPE and RMSE compared with the other two trials. However, the prediction of the API values with one hidden layer is not accurate when looking at the trend of the predicted API with the actual API.

The API values that are predicted by each of the trials are plotted in Fig. 3, and it can be clearly observed that the prediction of the first and second trial of the API values is stationary over the year. This makes no sense since it is impossible to have the same API readings over an entire year in real life. Therefore, comparing the third trial with the other two trials, the prediction of the third trial of the ANN model produces the best results among the three trials.

Therefore, the predicted API values obtained from the third trial of the ANN method with three layers of hidden layers will be compared against the predicted API values obtained via the ARIMA model and the three fuzzy time series models. The neural network of the third trial in the ANN model is plotted and shown in Fig. 4.

Analysis of results for the ARIMA model

The forecasted API values from the ARIMA(6, 1, 5) model is plotted in Fig. 5.

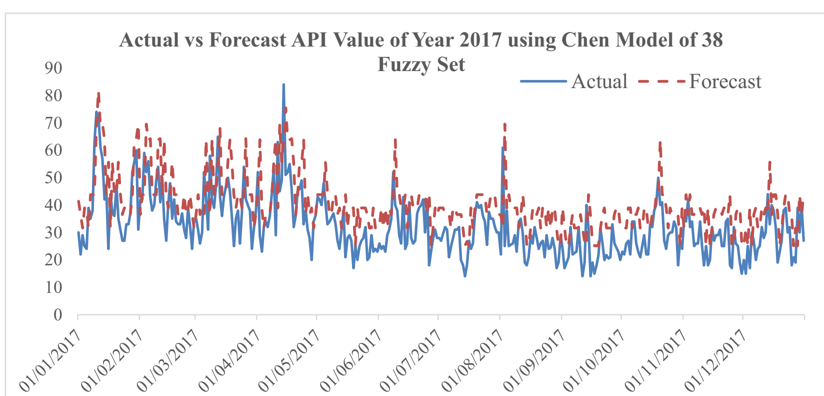

Fig. 7 Actual API vs forecasted API from Chen (1996) model

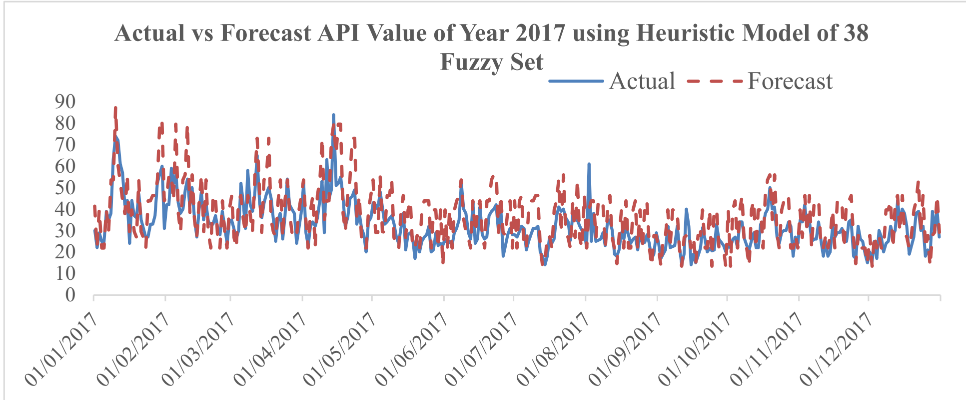

Fig. 8 Actual API vs forecasted API from the heuristic model (Huarng 2001)

Figure 5 shows that the forecasted values are increasing at a decreasing rate from the start. Later, it converges to the value of 51 towards the end of 2017. The forecasted value might not give any useful information as it does not really reflect the nature of API. The forecasts might not seem reasonable as the API would probably not converge to a constant level as shown in Fig. 5. Therefore, the ARIMA(6, 1, 5) might not be a suitable model to predict the API values. The model's forecast is further evaluated using the RMSE and MAPE as shown in Table 2

Analysis of results for the TBATS model

The forecasted API value from the TBATS (1, {1,3}, -, {<7,2>, <30,4>, <90,1>, <365,5>}) model is plotted in Fig. 6.

Based on the chart above, the TBATS model can generate forecasted values with seasonality similar to the actual API readings. This is because the TBATS model has applied complex seasonal differencing of weekly, monthly, trimonthly and annually to the time series. Thus, intuitively, the TBATS model provides a better forecast than the ARIMA model.

The MAPE and RMSE for the ARIMA and TBATS models are given in Table 2.

From Table 2, it can be seen that both the ARIMA(6, 1, 5) and TBATS models do not produce very accurate results in predicting the API values for the year 2017. In particular, the MAPE for the ARIMA(6, 1, 5) reaches as high as 67.595%. This might be due to the characteristics of the ARIMA model that is more suitable to explain and predict linear models. However, both models did provide a sensible computational time whereby the TBATS model requires a longer computational time compared with the nonseasonal ARIMA model due to the calculation needed for the complex seasonal differencing as well as the extra coefficient in the seasonal components. Thus, the forecasting results show that the API can be fitted better using a nonlinear model.

Analysis of results for the fuzzy time series models

The plots for the forecasted API values for the fuzzy time series models are presented in this section. The results of the forecasted API values vs the actual API values for the Chen (1996) FTS model, the heuristic FTS model (Huarng 2001) and the Singh (2008) FTS model are presented in Figs. 7, 8 and 9, respectively.

Table 3 tabulates the first 20 days of the API forecasts by the Chen (1996), heuristic (Huarng 2001) and Singh (2008) fuzzy time series models.

The first forecast produced by the Chen (1996) and heuristic (Huarng 2001) models would often be the same as both these methods share the same underlying rules for forecasting.

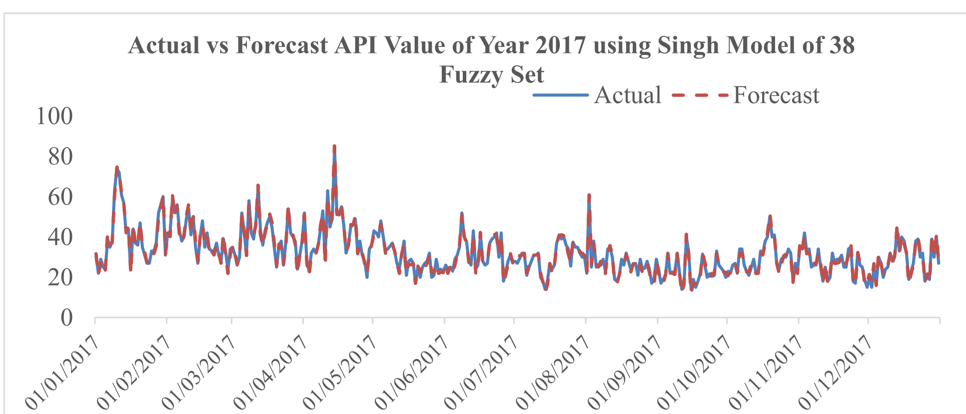

Fig. 9 Actual API vs forecasted API from the Singh (2008) model

Table 3 Comparison of forecasted API of the Chen, heuristic and Singh FTS models

Day	Actual	Forecasted by Chen method	Forecasted by the heuristic method	Forecasted by Singh method
01/01/2017	30	41.38158	41.38158	31.64474
02/01/2017	22	36.51316	21.90789	21.45395
03/01/2017	29	31.64474	39.55592	26.64766
04/01/2017	25	38.94737	21.90789	27.87829
05/01/2017	24	38.94737	21.90789	22.91447
06/01/2017	39	31.64474	39.55592	41.38158
07/01/2017	35	43.81579	29.21053	36.22661
08/01/2017	38	43.81579	53.55263	37.03216
09/01/2017	63	43.81579	53.55263	60.85526
10/01/2017	74	69.51023	87.41029	74.73026
11/01/2017	72	81.82692	60.85526	70.91959
12/01/2017	61	68.85338	52.94408	61.92763
13/01/2017	57	69.51023	43.81579	56.24342
14/01/2017	42	64.20230	37.90414	42.19079
15/01/2017	44	43.81579	55.98684	45.58333
16/01/2017	24	55.68257	29.55827	21.90790
17/01/2017	44	31.64474	39.55592	46.25000
18/01/2017	37	55.68257	29.55827	36.51316
19/01/2017	36	43.81579	26.77632	35.67325
20/01/2017	47	43.81579	53.55263	47.12500

Yet, the forecasting result would be different starting from the second period as the heuristic method depends on the increasing or decreasing trend of the data to determine the value of the next forecast.

Also, it is clear that the forecasted value from the Singh (2008) model is able to better predict the API values, and Chen (1996) model produces forecasted value that is most deviated from the actual value. The RMSE and MAPE values for all three FTS models are given in Table 4.

Table 4 clearly shows that the forecasted values obtained through the Singh (2008) model have the lowest MAPE and RMSE among the fuzzy time series models. Although Singh (2008) model is not the fastest in terms of computational time, it surpasses the other two models by having a reduction of at least 85% of the RMSE. This huge improvement in accuracy more than compensates for the slightly longer computation time for the Singh model.

The results show that the time-variant parameter, D_i, used by the Singh (2008) model can help to significantly boost the accuracy and efficiency of the model. The high-level accuracy of the Singh (2008) model in which the time-variant parameter is computed by the absolute differences of the past three lags of the data as the current state implies that the API at time i greatly depends on the API value of the past several lags.

Comparative studies

Throughout this section, a comparative analysis of the methods used in this research is presented. Six methods will be used for this purpose, namely the ANN model, ARIMA model by Box and Jenkins, TBATS model by De Livera et al. (2010), and the Chen (1996), heuristic (Huarng 2001) and Singh (2008) fuzzy time series models. All these models will be applied to forecast the API value of Kuala Lumpur for the year 2017. The output of each model will be evaluated using the error metrics of RMSE and MAPE to verify its accuracy.

Comparison of outcomes generated using different models

Discussion of results

Based on the results shown in Table 5, the fuzzy time series methods have, on average, lower RMSE and MAPE compared with the other models that were studied. This is mainly due to the ability of the FTS models to accurately capture the fluctuations and seasonality of the API readings, which is bound to see some drastic changes due to changes in the weather and changes in human activities during different periods in the city. Among all the models that were compared, the Singh (2008) FTS model has produced the forecasted values with the lowest RMSE and MAPE, and also has the second fastest computation time needed to produce the output, compared with all of the studied methods. Although the Singh

Table 4 MAPE and RMSE of the three FTS models

FTS model	Chen	Heuristic	Singh
MAPE	36.356%	26.518%	4.364%
RMSE	12.37937	10.09388	1.47040
Computational time (min)	2.21	1.48	1.53

Table 5 The results obtained using the chosen six methods

Model	Specific model used	RMSE	MAPE	Time taken (min)
ANN	1 Hidden Layer	22.21475	78.781%	2.17
	2 Hidden Layers	22.36465	79.051%	2.32
	3 Hidden Layers	22.72587	78.736%	3.03
ARIMA	ARIMA(6, 1, 5)	19.89427	67.595%	2.02
TBATS	TBATS (1, {1,3}, -, {<7,2>, <30,4>,<90,1>, <365,5>})	15.15626	43.861%	2.13
FTS	Chen (1996) model	12.37937	36.356%	2.21
	Heuristic model (Huarng 2001)	10.09388	26.518%	1.48
	Singh (2008) model	1.47040	4.364%	1.53

(2008) model might not be the fastest in terms of computational time, it surpasses the other two FTS models by recording an RMSE that is almost 85% lower compared with the other two FTS models. This huge improvement in accuracy more than compensates for the 0.05-min delay in computation time for this model compared with the heuristic model (Huarng 2001). Hence, we can conclude that the Singh (2008) FTS model is the most accurate and efficient model to predict the Air Pollution Index (API) of Kuala Lumpur, Malaysia, compared with the other models that were studied. As the Singh (2008) FTS model works by predicting the API values for the next state using a time-variant parameter D_i that is computed by taking the absolute differences of the past three lags of the data as the current state, it is particularly useful in forecasting situations in which the current values or readings are dependent on the values or readings of the past several lags. This explains the accuracy of the results produced by the Singh (2008) FTS model.

Conclusion

The concluding remarks, as well as the limitations of the work done in this paper, are summarized below:

i. The models used in this research include the trigonometric regressors, Box-Cox transformation, ARMA errors, trend and seasonality (TBATS) model, artificial neural network (ANN) model, autoregressive integrated moving average (ARIMA) model and three fuzzy time series (FTS) models. All these models were used to predict the Air Pollution Index (API) readings for Kuala Lumpur, Malaysia, in the year 2017, using actual datasets of the API readings of Kuala Lumpur from the year 2012 to 2017. The data for the year 2012–2016 were used as training data, whereas the data for year 2017 were used as testing data.

ii. The performance of the chosen models in forecasting the API values were evaluated using two error metrics, namely the root mean squared error (RMSE) and mean absolute percentage error (MAPE), and the computation time. Among all of the models that were studied, it was found that the Singh (2008) FTS model produced the most accurate forecasted values with the lowest value of RMSE of 1.4704 and the lowest value of MAPE of 4.364%. It is also the second fastest model in terms of computation time second only to the heuristic FTS model by Huarng (2001). This led us to conclude that the Singh (2008) FTS model is the most accurate and efficient model compared with all of the chosen models to predict the API values of Kuala Lumpur for the year 2017.

iii. One of the main limitations that we faced in this study is handling the 91 missing historical API data points when the dataset was retrieved from the Malaysia Public Sector Open Data Portal. The reason for the missing data remains unidentified. While the missing data represents only 4% of the data from the year 2012 to 2017, some assumptions were made in handling these missing data. As these data were all missing between 2 days with similar or almost similar API values, a linear interpolation was made between the API readings of the 2 days (before and after) to obtain the missing data. Hence, this assumption might have a minor effect on the results that were obtained.

iv. The results obtained clearly indicate that the fuzzy time series model is able to forecast the API values accurately. This accurate method of API forecasting can be utilized by the relevant government agencies to better plan and initiate the necessary preventive measures or mitigation plans to reduce the impact of air pollution on the people. These forecasts will also help government departments and agencies to plan outdoor activities and outreach programmes during periods of lower API, and inform citizens to reduce outdoor activities and wear safety masks during periods of higher API. These are important to protect the peoples' health from being affected by air pollution.

v. In the future, we will try to overcome these limitations using new forecasting methods in existing literature (Son and Fujita 2019; Tuan et al. 2019; Son et al. 2019; Hoang et al. 2019; Jha et al. 2019; Binh et al. 2018; Binh et al. 2019; Pham et al. 2018; Tuong et al. 2018; Jain et al. 2018).

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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